

# Practice Problem Set 1

ECON 308 – International Economic Relations

Fall 2025

1. Explain why country size (GDP) and geographic distance are strong predictors of bilateral trade flows. Provide at least one microeconomic channel for each.
2. In the Ricardian model with free trade, what pins down relative wages across countries? Explain the role of world prices and specialization.
3. Briefly explain why, under perfect competition, autarky relative prices equal unit labor cost ratios, and how opening to trade changes the mapping from technology to prices and wages.
4. The (log) gravity model is

$$\ln T_{ij} = \alpha + \beta_1 \ln Y_i + \beta_2 \ln Y_j - \gamma \ln D_{ij},$$

with  $T_{ij}$  bilateral exports from  $i$  to  $j$ ,  $Y$  GDP, and  $D$  great-circle distance (km). Suppose  $\beta_1 = \beta_2 = 1$ ,  $\gamma = 1$ , and  $e^\alpha = 10^{-6}$  so that

$$T_{ij} = 10^{-6} \frac{Y_i Y_j}{D_{ij}}.$$

Consider the following economies (values in billions of US dollars; distances in km):

| Country | $Y$ (\$ bn) |     |       |       |
|---------|-------------|-----|-------|-------|
|         |             | $A$ | $B$   | $C$   |
| $A$     | 2,000       | $A$ | –     | 6,000 |
| $B$     | 1,000       | $B$ | 6,000 | –     |
| $C$     | 500         | $C$ | 1,500 | 2,000 |

- (a) Compute the predicted bilateral exports  $T_{AB}$ ,  $T_{AC}$ , and  $T_{BC}$ .
- (b) By what *percent* does  $T_{AB}$  change if  $D_{AB}$  falls by 10%? (Use the elasticity implied by the model.)
- (c) If both  $Y_A$  and  $Y_B$  rise by 5%, what is the approximate percentage change in  $T_{AB}$ ?
- (d) Suppose the true  $T_{AB}$  observed in data is \$1.30 billion. Compute the (log) residual  $\ln T_{AB}^{\text{obs}} - \ln T_{AB}^{\text{pred}}$  and provide one economic interpretation of a positive residual.

5. Two countries, Home ( $H$ ) and Foreign ( $F$ ), produce Wine ( $W$ ) and Cloth ( $C$ ) with one input, labor. Unit labor requirements are:

|     | $a_{LW}$ | $a_{LC}$ |                                               |
|-----|----------|----------|-----------------------------------------------|
| $H$ | 2        | 4        | Labor endowments: $L_H = 240$ , $L_F = 240$ . |
| $F$ | 6        | 3        |                                               |

- Compute each country's autarky opportunity cost of  $W$  in terms of  $C$  and vice versa. Identify comparative advantage.
  - Draw (or describe algebraically) each country's PPF. What is each country's autarky relative price  $P_W/P_C$  under competitive pricing equal to unit labor cost?
  - Find the range of world relative prices  $P_W/P_C$  consistent with complete specialization.
  - Suppose the world relative price is  $P_W/P_C = 1.2$ . Determine specialization, production bundles, and the pattern of trade.
  - Compute the real wage in each country under free trade in terms of each good (units of  $W$  and units of  $C$  purchasable per unit of labor income).
6. Continue from previous question with  $P_W/P_C$  moving from 1.2 to 0.8.
- Re-evaluate which country (if any) changes specialization as the relative price falls to 0.8. Explain.
  - Compute the real wage of each country in terms of  $W$  and  $C$  at  $P_W/P_C = 0.8$  and compare to autarky.
  - Which country's real wage benefits from an improvement in its terms of trade? Explain using the Ricardian logic.
7. There are three countries with the following unit labor requirements:

|   | $a_{LW}$ | $a_{LC}$ |                           |
|---|----------|----------|---------------------------|
| 1 | 1        | 2        | $L_1 = L_2 = L_3 = 100$ . |
| 2 | 2        | 3        |                           |
| 3 | 4        | 5        |                           |

- Order countries by comparative advantage in  $W$  relative to  $C$ .
- Derive the world relative supply (WRS) schedule  $RS^{W/C}(P_W/P_C)$  by progressively "turning on" countries into  $W$  as  $P_W/P_C$  rises. State the price thresholds where kinks occur.
- If world relative demand is  $RD^{W/C} = 1.1 - 0.2(P_W/P_C)$  (for  $P_W/P_C \in [0, 5]$ ), solve for the free-trade equilibrium  $P_W/P_C$  and identify which countries produce which good.